Automotive Technical Service Diagnostic Bulletin

Document ID: DIAG-S-CLASS-P0113
Classification: OEM Level 3 Diagnostic & Repair Procedure
Target Vehicle: Mercedes-Benz S-Class (W223 / S 500 4MATIC)
Powertrain: 3.0L Turbocharged Inline-6 (M256) with 48V Integrated Starter-Generator (ISG) & Electric Auxiliary Compressor (eZV)
Engine Control System: Bosch MG1CP002 (Engine Control Module N3/10) - FlexRay / CAN-FD Gateway
Electrical Architecture: Dual 12V / 48V Mild-Hybrid Architecture
Fuel Injection Architecture: 200-350 bar Direct Injection (Piezo HP Fuel Injectors, dual HP fuel pumps)
Firing Order: 1-5-3-6-2-4

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1. DIAGNOSTIC TROUBLE CODE (DTC) SPECIFICATION

- **DTC Code:** `P0113`
- **Standard Fault Description:** Intake Air Temperature (IAT) Sensor 1 Circuit High Input
- **System Definition:** IAT thermistor circuit voltage exceeds 4.85V, indicating an open circuit or unplugged harness connector.
- **MIL (Malfunction Indicator Lamp) Status:** Illuminated (Continuous or Flashing depending on catalyst severity)
- **Monitoring Condition:** Continuous monitoring during engine running cycle with coolant temp > 60°C and vehicle speed > 0 km/h.

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2. PLATFORM-SPECIFIC ROOT CAUSE ANALYSIS (RCA)

Root cause analysis customized specifically for the **Mercedes-Benz S-Class (W223 / S 500 4MATIC)**:

1. **Sensor Internal Thermistor Open Circuit:** Negative Temperature Coefficient (NTC) bead fractured internally.
2. **Harness Ground Return Broken:** Pin 1 ground circuit wire sheared inside corrugated conduit near front intercooler charge pipe.
3. **Connector Pin Terminal Spread:** Female terminal backing out of connector housing B17/9 due to aggressive previous back-probing.

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3. DEDICATED WIRING DIAGRAM & CIRCUIT PINOUT

Schematic configuration and electrical pin connections specifically for the S-Class control harness:

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[IAT Sensor B17/9]                          [ECM N3/10 Connector 'E']
Pin 1 (Low Reference / Signal Return) ----> Pin 41 (Sensor Ground)
Pin 2 (5.0V Pull-Up Signal Line) <-------- Pin 42 (5V Pull-up / ADC Input)
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- Voltage Characteristic: At 20°C: ~2.5kΩ (approx 2.4V); At 80°C: ~300Ω (approx 0.8V); Open Circuit (>4.85V = -40°C indicated).

Electrical Reference Points

- **Primary Ground Reference:** `W10/4 (Cylinder Head Ground, Left) & W16/5 (Front Bulkhead Ground)`
- **Power Distribution Node:** `Front SAM (Signal Acquisition Module N10/6) Fuse Box F32/3`
- **Signal Logic Voltage Range:** 0.0V to 5.0V regulated DC (unless high-voltage direct injection rail)

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4. STEP-BY-STEP DIAGNOSTIC & VERIFICATION PROCEDURE

Follow this OEM workshop sequence to isolate the failure on this platform:

1. View live IAT reading with scan tool: if displaying -40°C (-40°F), open circuit is confirmed.
2. Disconnect sensor B17/9; bridge Pin 1 and Pin 2 with a 100Ω resistor: if scan tool reads ~120°C, ECU and harness are intact (defective sensor).
3. If scan tool still reads -40°C with jumper, measure continuity between Pin 1 and Chassis Ground, and Pin 2 to ECU Pin 42.
4. Check terminal retention tension using terminal drag test tool.

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5. PRECISE RESOLUTION & REPAIR PROCEDURE

- Replace IAT Sensor B17/9 (screw-in / clip-in sensor on charge air pipe).
- Re-pin connector terminal if loose using genuine Mercedes MCON 1.2 gold-plated contact pins.
- Clear stored and freeze-frame fault codes in N3/10.

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6. POST-REPAIR QUALITY VALIDATION & RE-TESTING

1. Clear all DTCs from Engine Control Module (`Bosch`) and secondary gateway modules.
2. Perform comprehensive drive cycle:
3. Cold start idle for 3 minutes (monitors secondary air / catalyst preheat and cold misfire).
4. Accelerate smoothly to 50 km/h and hold for 2 minutes.
5. Cruise at 80–100 km/h on highway for 10 minutes (validates downstream O2 sensor and fuel trim closed-loop balance).
6. Coasting deceleration (fuel cut-off check) for 5 seconds without braking.
7. Re-scan control units with diagnostic tool; confirm all OBD-II readiness monitors show **READY** / **COMPLETE** and no pending DTCs are stored.